BUILDING AI INFRASTRUCTURE · PART 6 OF 7

Part 6: On-Premises — Your Data Center

Chapter summary for beginners

In one sentence: leave the cloud and run the same platform on hardware you own. The top layers stay the same, and every layer underneath becomes your job.

Video length: about 8 minutes · **Code:** nawex-platform `deploy/ansible`, openshift-k8s-ai-infra `platform/openshift`, this repo's `flux/clusters/onprem-openshift` · **Cost:** hardware you already own

Why AI runs on-prem

Banks, hospitals, defense and government often can't or won't send data to a public cloud. And if GPUs run all day, every day, owning them can cost less than renting them.

The 8 blocks on-prem

#	Block	On your own hardware
1	Compute	GPU servers (for example NVIDIA HGX-class) + CPU servers
2	Data and storage	**Ceph** / OpenShift Data Foundation, with **MinIO** for an S3-compatible API
3	Networking	VLANs, firewall, internal DNS, **MetalLB** (load balancer IPs on bare metal)
4	Orchestration	**OpenShift** (or RKE2 / kubeadm)
5	MLOps	MLflow + Kubeflow, or OpenShift AI
6	Serving	KServe + vLLM on GPU worker nodes
7	Observability	Prometheus, Grafana, OpenTelemetry, often Splunk
8	Guardrails	Active Directory / LDAP, **Vault**, RBAC, OpenCost

The honest trade

Now your job	What you gain	What you trade
Buy, rack, power and cool servers	Data never leaves the building	No scaling to zero on hardware you own
Install and patch the OS and drivers	Fixed cost for GPUs running 24/7	Months to buy GPUs, not minutes
Networking and load balancers	Full control, air-gap options	You need the skills in-house
Storage, backups, spare parts	Meets strict regulators	Capacity planning matters

From bare metal to serving: 5 steps

1. **Baseline the OS** with Ansible: patching, time sync and hardening (nawex-platform has a RHEL baseline playbook).
2. **Install OpenShift**, or a lighter Kubernetes such as RKE2.
3. **Storage and network:** ODF/Ceph for storage, and MetalLB so services get load balancer IPs.
4. **GPU operators: Node Feature Discovery** first, then the **NVIDIA GPU Operator with the driver turned on**, because there's no cloud image to provide it.
5. **The platform itself** with Helm: MLflow, KServe and vLLM. In this repo, `make flux-bootstrap CLUSTER=onprem-openshift` adds the cluster to the GitOps fleet.

Air-gapped and regulated

Many on-prem sites have **no internet at all**:

- Mirror container images to an internal registry.
- Mirror Helm charts and OS packages too.
- Bring model weights in through a reviewed, approved path.

Regulators need proof:

- Every change goes through Git and a review.
- Audit logs are kept and protected from editing.
- Access is tied to named people, not shared accounts.

> If you can't show who changed what, and when, it didn't pass the audit.

Capacity: you can't scale to zero

The hardware is there whether you use it or not, so the goal is to use every GPU well:

- **MIG** splits one big GPU into isolated slices for several small models.
- **Time-slicing** shares a GPU for notebooks and light work.
- **Quotas per team** stop one group from taking every GPU.
- **Measure weekly** with DCGM metrics. This repo's `GPUWeeklyUtilizationLow` alert does it for you.

Words to know

- **Bare metal:** physical servers with no cloud layer.
- **OpenShift:** Red Hat's enterprise Kubernetes distribution.
- **MetalLB:** gives Kubernetes load balancer addresses on bare metal.
- **Node Feature Discovery (NFD):** detects hardware such as GPUs and labels the nodes.
- **Air-gapped:** fully disconnected from the internet.

- **MIG:** Multi-Instance GPU, one physical GPU split into isolated slices.

Check yourself

1. Why is the GPU Operator's driver turned **on** here but off in the clouds?
2. What does MetalLB replace?
3. Name two things you must mirror for an air-gapped install.
4. Why measure GPU usage every week on-prem?

Answers: 1) No cloud image provides the driver on your own servers. 2) The cloud's managed load balancer. 3) Any two of: container images, Helm charts, OS packages, model weights. 4) Buying more GPUs takes months, so you have to see shortages coming.

Next: Part 7, the finale: one platform across all four.

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Code: github.com/Here2ServeU/building-ai-infra · Series guide: docs/series/README.md